

KON 313

Feedback Control Systems

Modeling in the Time Domain

Dr. Aysa Jafari Farmand

aysa.jafari@itu.edu.tr

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Introduction

Two approaches of analysis and design of feedback control systems:

1. Frequency-domain (classical) technique

- can be applied only to linear, time-invariant systems.

- + rapidly provide stability and transient response information.

2. The state-space (modern, or time-domain) approach

- + can represent wide range of systems, such as nonlinear systems, systems with nonzero initial conditions, time-varying systems, Multiple-input multiple-output systems

- is not as intuitive as the classical approach.

Introduction

1. We select a particular *subset* of all possible system variables and call the variables in this subset *state variables*.
2. For an n th-order system, we write n *simultaneous, first-order differential equations* in terms of the state variables. We call this system of simultaneous differential equations *state equations*.
3. If we know the initial condition of all of the state variables at t_0 as well as the system input for $t \geq t_0$, we can solve the simultaneous differential equations for the state variables for $t \geq t_0$.
4. We *algebraically* combine the state variables with the system's input and find all of the other system variables for $t \geq t_0$. We call this algebraic equation the *output equation*.
5. We consider the state equations and the output equations a viable representation of the system. We call this representation of the system a *state-space representation*.

State-space Representation

Some definitions:

- *Linear combination.* A linear combination of n variables, x_i , for $i = 1$ to n , is given by

$$S = K_n x_n + K_{n-1} x_{n-1} + \cdots + K_1 x_1$$

where each K_i is a constant.

- *Linearly Independence.* Variables x_i , for $i = 1$ to n , are said to be linearly independent if their linear combination, S , equals zero only if every $K_i = 0$ and no $x_i = 0$ for all $t \geq 0$.

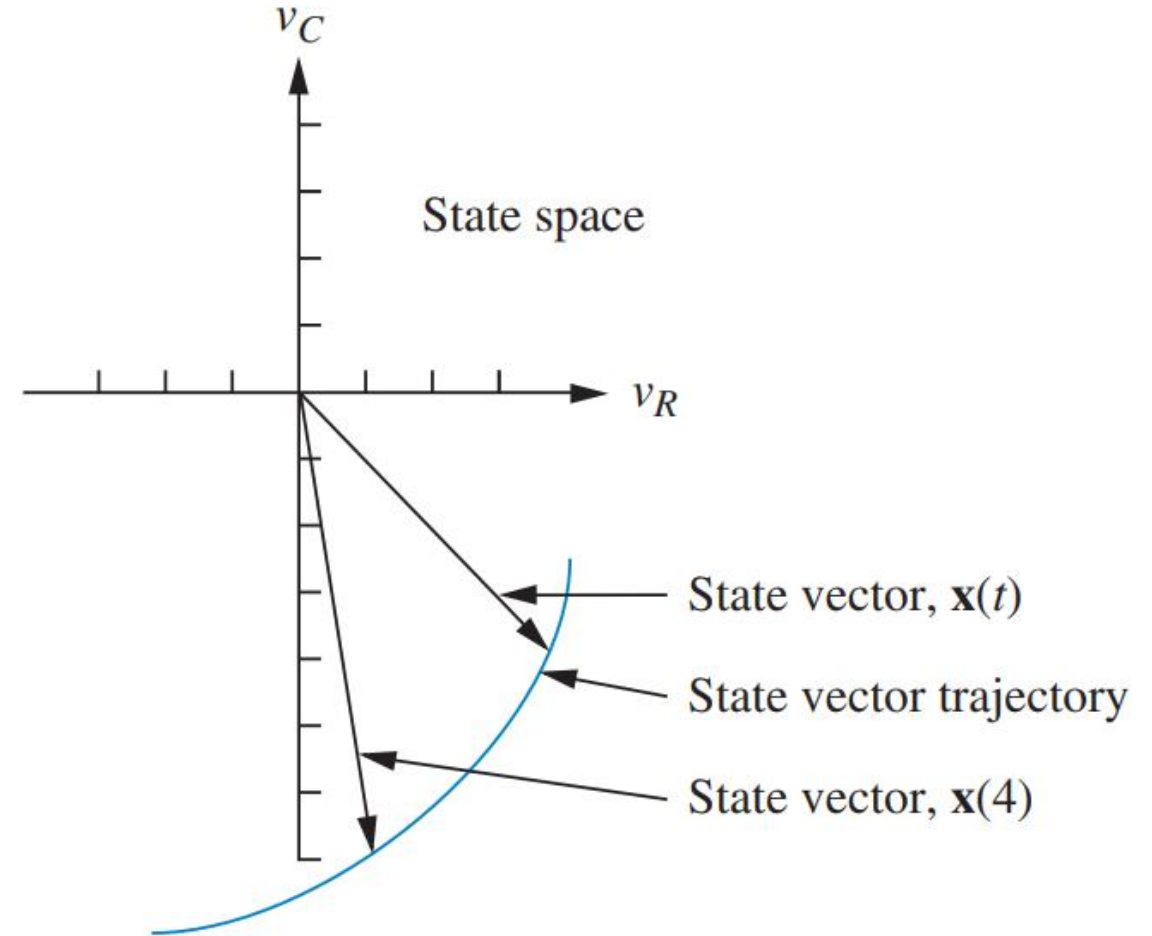
State-space Representation

Some definitions:

- *System variable*. Any variable that responds to an input or initial conditions in a system.
- *State variables*. The smallest set of linearly independent system variables such that the values of the members of the set at time t_0 along with known forcing functions completely determine the value of all system variables for all $t \geq t_0$.
- *State vector*. A vector whose elements are the state variables.

State-space Representation

- *State space.* The n -dimensional space whose axes are the state variables. A trajectory can be thought of as being mapped out by the state vector, $\mathbf{x}(t)$, for a range of t .
- *State equations.* A set of n simultaneous, first-order differential equations with n variables, where the n variables to be solved are the state variables.
- *Output equation.* Algebraic equation that expresses the output variables of a system as linear combinations of the state variables and the inputs.



State-space Representation

A system is represented in state space by the following equations:

for $t \geq t_0$ and initial conditions, $x(t_0)$

$\mathbf{x}$ = state vector

$\dot{\mathbf{x}}$ = derivative of the state vector with respect to time

$\mathbf{y}$ = output vector

$\mathbf{u}$ = input or control vector

$\mathbf{A}$ = system matrix

$\mathbf{B}$ = input matrix

$\mathbf{C}$ = output matrix

$\mathbf{D}$ = feedforward matrix

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

Applying the State-space Representation

- It should be noted that, the state-space representation is not unique.

The first step in representing a system is to select the state vector, which must be chosen according to the following considerations:

1. A minimum number of state variables must be selected as components of the state vector. This minimum number of state variables is sufficient to describe completely the state of the system.
 - Typically, the minimum number required equals the order of the differential equation describing the system.
2. The components of the state vector (that is, this minimum number of state variables) must be linearly independent.
 - Variables and their successive derivatives are linearly independent.

Applying the State-space Representation

Representing an Electrical Network (An alternative approach):

Step 1. Write the simple derivative equation for each **energy-storage element**, and select each differentiated variable as a state variable.

Step 2. Solve for each derivative term as a linear combination of any of the state variables and the input that are present in the equation.

Step 3. Write the output variables as linear combinations of the state variables and the input

Applying the State-space Representation

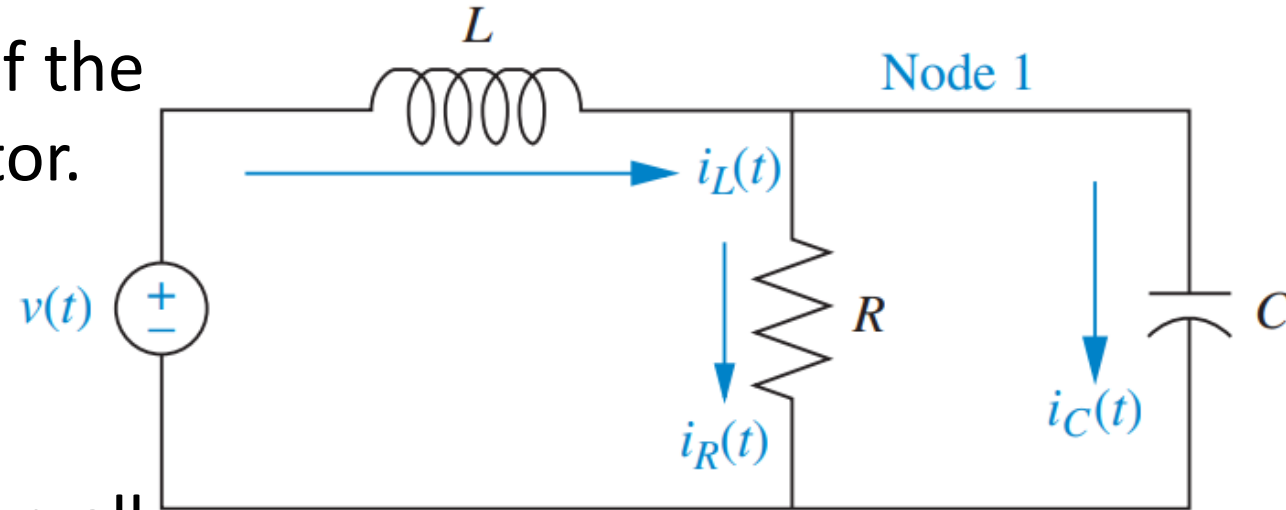
E.g. Find a state-space representation if the output is the current through the resistor.

Sln.

1. Label all of the branch currents.
2. Writing the derivative equation for all energy-storage elements.

$$C \frac{dv_C}{dt} = i_C \quad L \frac{di_L}{dt} = v_L$$

v_C and i_L are chosen as the state variables



3. KCL at Node 1

$$i_C = -i_R + i_L$$

$$= -\frac{1}{R}v_C + i_L$$

KVL around the outer loop

$$v_L = -v_C + v(t)$$

Applying the State-space Representation

E.g. Find a state-space representation if the output is the current through the resistor.

Sln (Cont.).

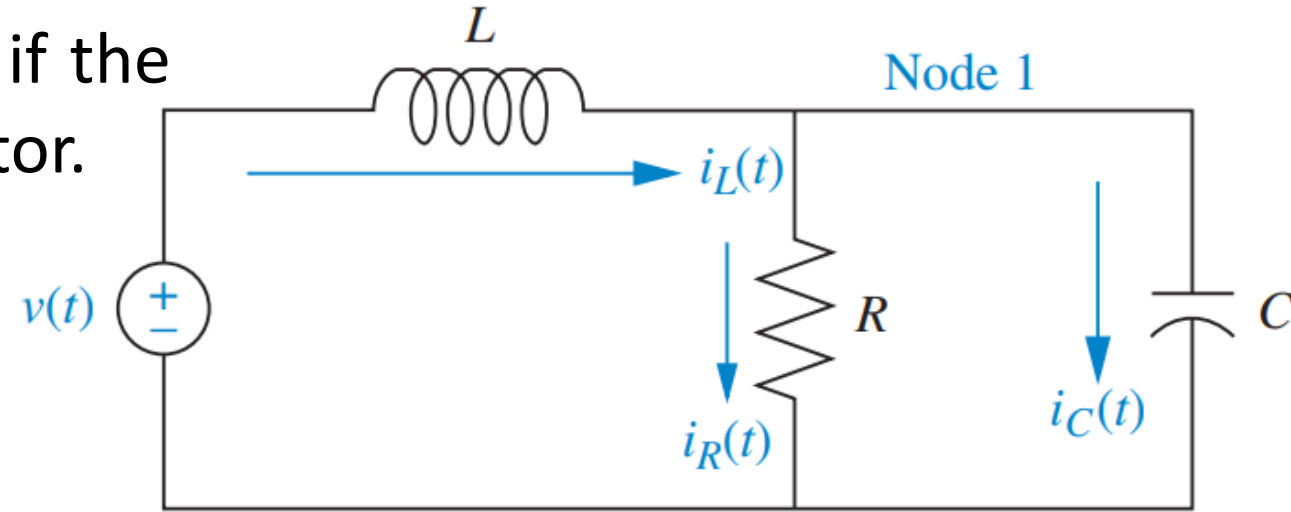
4. Substituting:

$$\frac{dv_C}{dt} = -\frac{1}{RC}v_C + \frac{1}{C}i_L$$

$$\frac{di_L}{dt} = -\frac{1}{L}v_C + \frac{1}{L}v(t)$$

5. Output equation

$$i_R = \frac{1}{R}v_C$$



The state-space representation

$$\begin{bmatrix} \dot{v}_C \\ \dot{i}_L \end{bmatrix} = \begin{bmatrix} -1/(RC) & 1/C \\ -1/L & 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} v(t)$$

$$i_R = \begin{bmatrix} 1/R & 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix}$$

Applying the State-space Representation

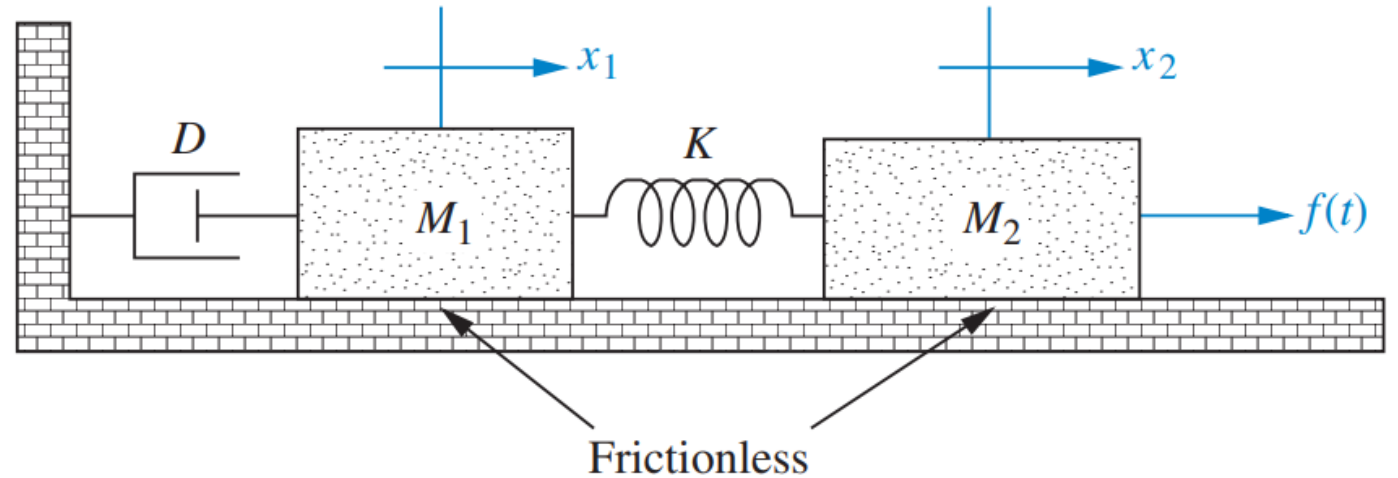
Representing a translational mechanical system (An alternative approach):

It is more convenient when working with mechanical systems to obtain the state equations directly from the equations of motion rather than from the energy-storage elements.

In mechanical systems we change our selection of state variables to be the **position and velocity of each point of linearly independent motion**.

Applying the State-space Representation

E.g. Find the state equations.



Sln.

1. write the differential equations for the network. (You can find the Laplace-transformed equations of motion, and next take the inverse Laplace transform of these equations, assuming zero initial conditions)

$$M_1 \frac{d^2 x_1}{dt^2} + D \frac{dx_1}{dt} + Kx_1 - Kx_2 = 0$$

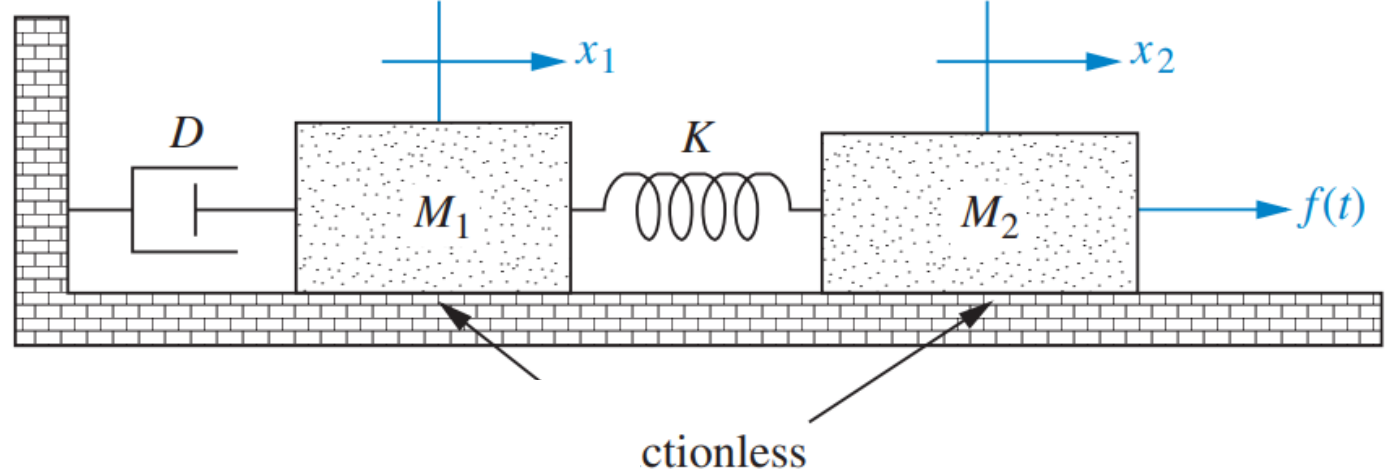
$$-Kx_1 + M_2 \frac{d^2 x_2}{dt^2} + Kx_2 = f(t)$$

Applying the State-space Representation

E.g. Find the state equations.

Sln (Cont.).

2. Select x_1, v_1, x_2, v_2 as state variables.



$$\frac{dx_1}{dt} = +v_1$$

$$\frac{dv_1}{dt} = -\frac{K}{M_1}x_1 - \frac{D}{M_1}v_1 + \frac{K}{M_1}x_2$$

$$\frac{dx_2}{dt} = +v_2$$

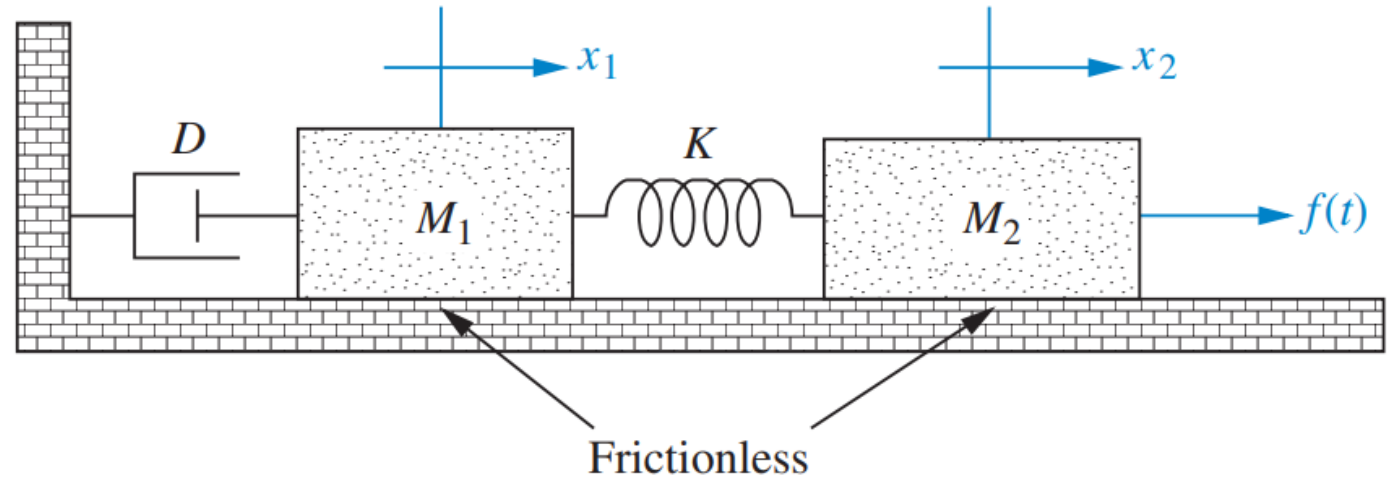
$$\frac{dv_2}{dt} = +\frac{K}{M_2}x_1 - \frac{K}{M_2}x_2 + \frac{1}{M_2}f(t)$$

Applying the State-space Representation

E.g. Find the state equations.

Sln (Cont.).

In vector-matrix form:



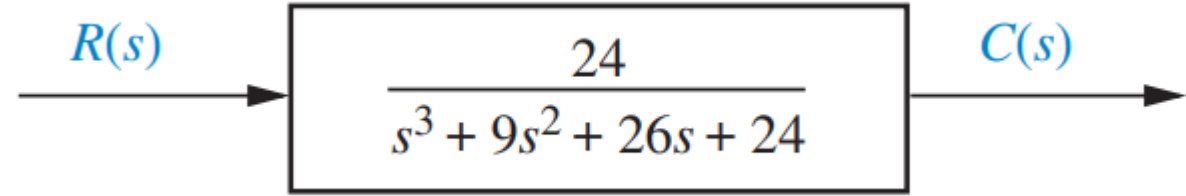
$$\begin{bmatrix} \dot{x}_1 \\ \dot{v}_1 \\ \dot{x}_2 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -K/M_1 & -D/M_1 & K/M_1 & 0 \\ 0 & 0 & 0 & 1 \\ K/M_2 & 0 & -K/M_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M_2 \end{bmatrix} f(t)$$

Converting a Transfer Function to State Space

Here, we select a set of state variables, called ***phase variables***, where each subsequent state variable is defined to be the derivative of the previous state variable.

Converting a Transfer Function to State Space

E.g. Converting a Transfer Function with a Constant Term in the Numerator



Step 1. Find the associated differential equation

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$
$$(s^3 + 9s^2 + 26s + 24)C(s) = 24R(s)$$
$$\ddot{c} + 9\ddot{c} + 26\dot{c} + 24c = 24r$$

Step 2. Select the state variables as successive derivatives.

$$x_1 = c$$

$$x_2 = \dot{c}$$

$$x_3 = \ddot{c}$$

Differentiating both sides and using the equation achieved in step 1

$$\dot{x}_1 = x_2$$

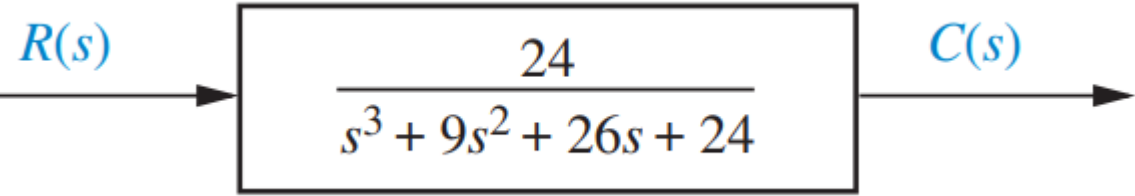
$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + 24r$$

$$y = c = x_1$$

Converting a Transfer Function to State Space

E.g. (Cont.) Converting a Transfer Function with a Constant Term in the Numerator



In vector-matrix format:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + 24r$$

$$y = c = x_1$$

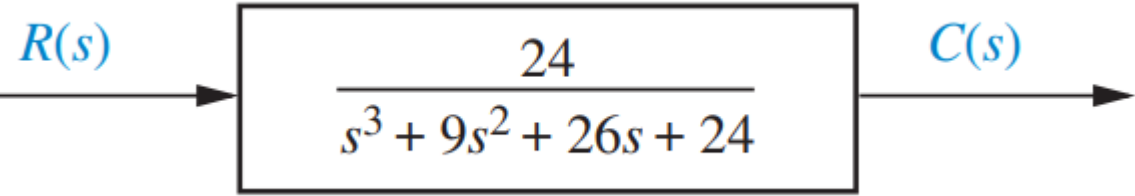
$\Rightarrow$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

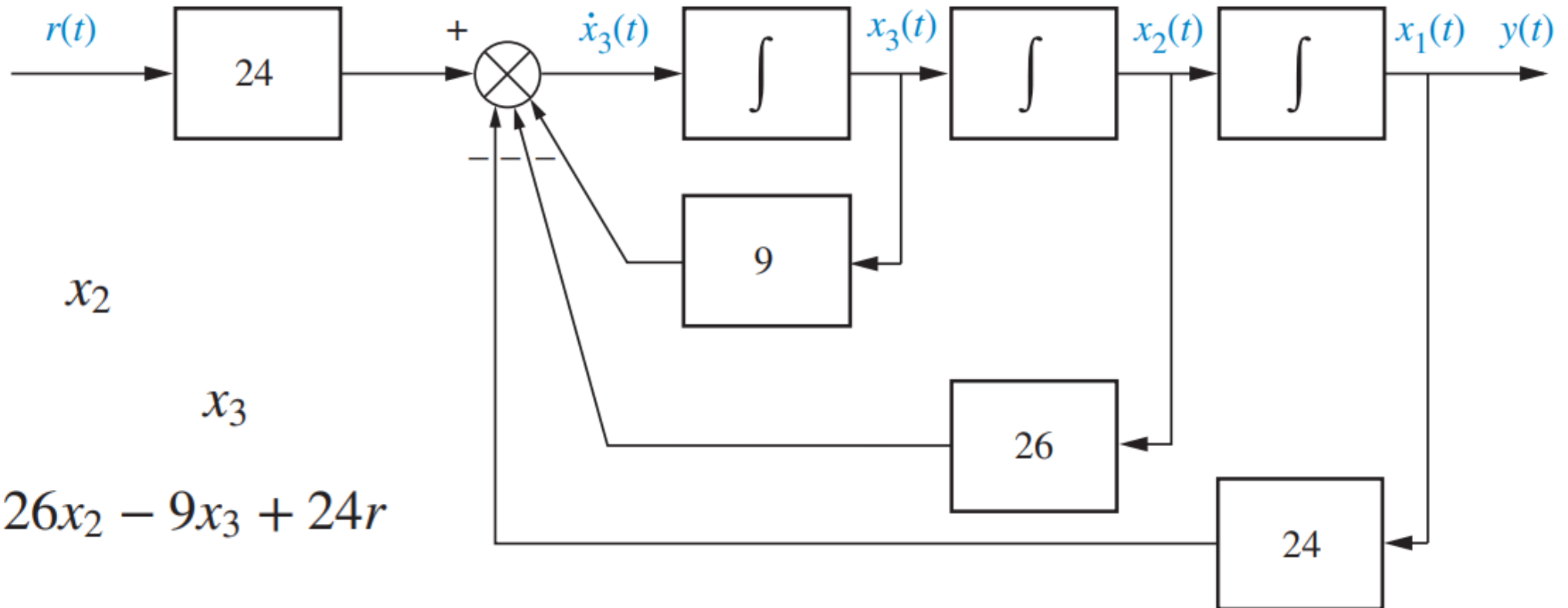
$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Converting a Transfer Function to State Space

E.g. (Cont.) Converting a Transfer Function with a Constant Term in the Numerator



In block-diag



$$\dot{x}_1 =$$

$$x_2$$

$$\dot{x}_2 =$$

$$x_3$$

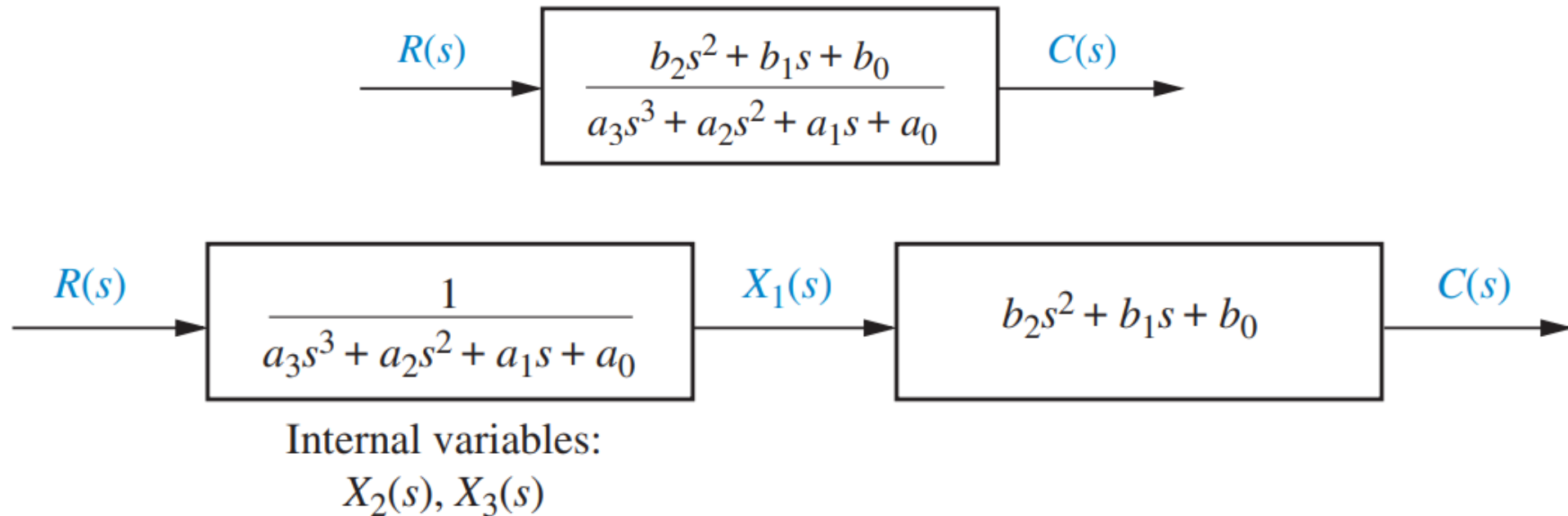
$$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + 24r$$

$$y = c = x_1$$

Converting a Transfer Function to State Space

Converting a Transfer Function with a Polynomial in the Numerator

- First separate the transfer function into two cascaded transfer functions; the first is the denominator, and the second is just the numerator.



- The denominator of the transfer function yields the state equations, while the numerator yields the output equation.

Converting a Transfer Function to State Space

Converting a Transfer Function with a Polynomial in the Numerator

- The first transfer function with just the denominator is converted to the phase-variable representation in state space, and the phase-variable x_1 is the output.
- The second transfer function with just the numerator yields

$$Y(s) = C(s) = (b_2s^2 + b_1s + b_0) X_1(s) \quad \xRightarrow{\mathcal{L}^{-1}} \quad y(t) = b_2 \frac{d^2x_1}{dt^2} + b_1 \frac{dx_1}{dt} + b_0x_1$$

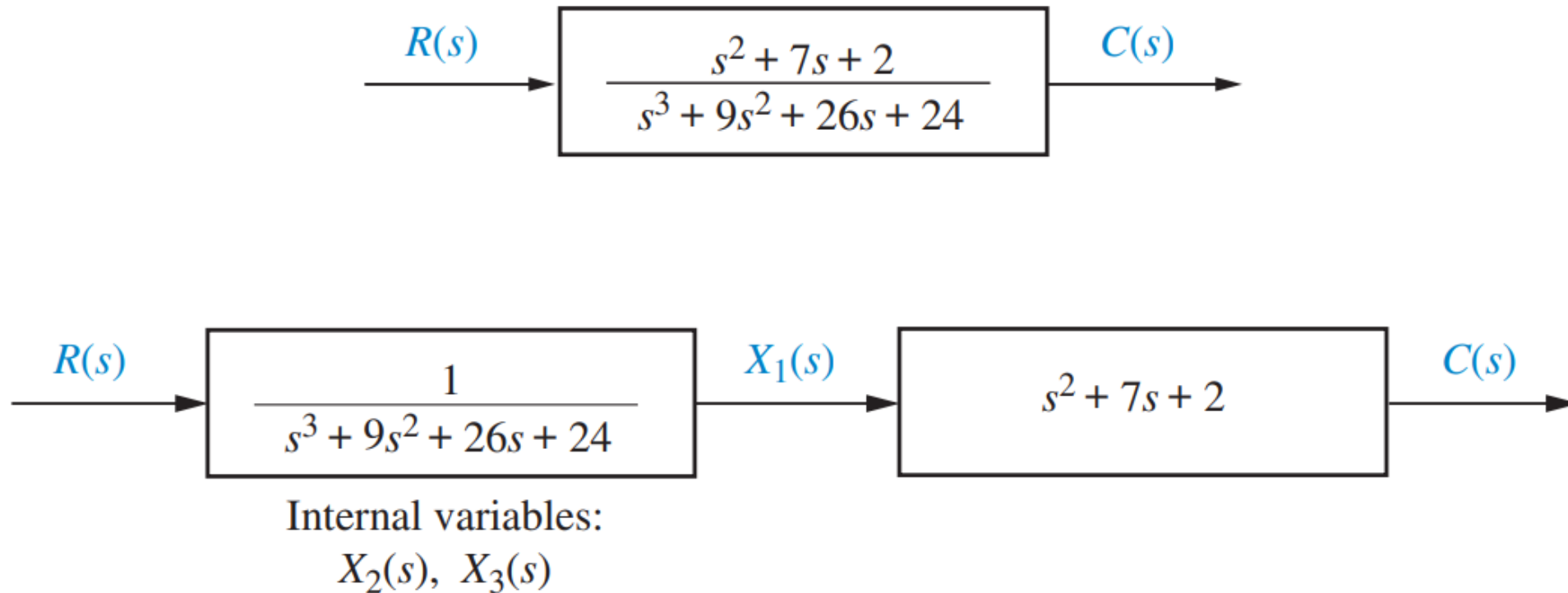
The derivative terms are the definitions of the phase variables obtained in the first block

$$y(t) = b_0x_1 + b_1x_2 + b_2x_3$$

Converting a Transfer Function to State Space

E.g. Converting a Transfer Function with a Polynomial in the Numerator

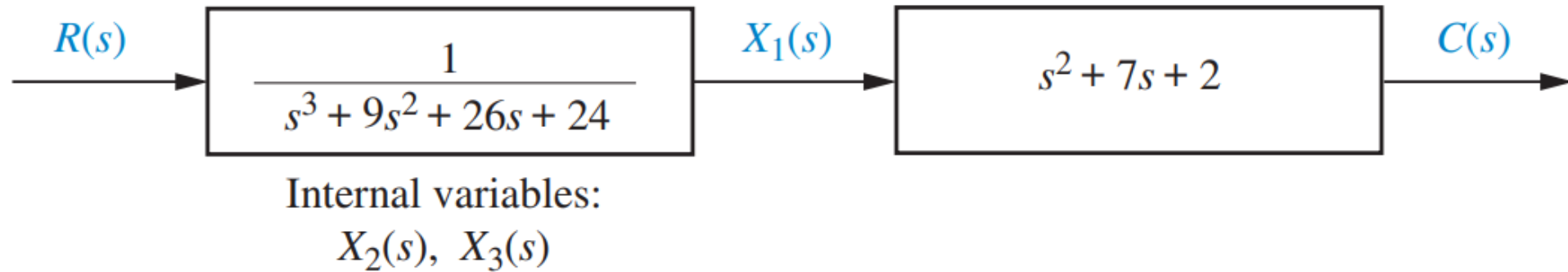
- Find the state-space representation of the shown transfer function.



Converting a Transfer Function to State Space

E.g. (Cont.) Converting a Transfer Function with a Polynomial in the Numerator

- Find the state-space representation of the shown transfer function.

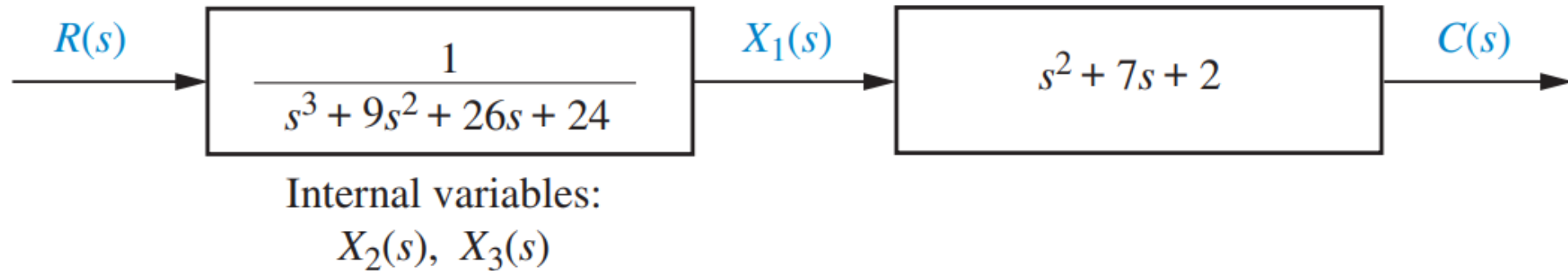


$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

Converting a Transfer Function to State Space

E.g. (Cont.) Converting a Transfer Function with a Polynomial in the Numerator

- Find the state-space representation of the shown transfer function.



$$\begin{aligned} C(s) &= (b_2 s^2 + b_1 s + b_0) X_1(s) \\ &= (s^2 + 7s + 2) X_1(s) \end{aligned}$$

$$c = \ddot{x}_1 + 7\dot{x}_1 + 2x_1$$

$$x_1 = x_1$$

$$\dot{x}_1 = x_2$$

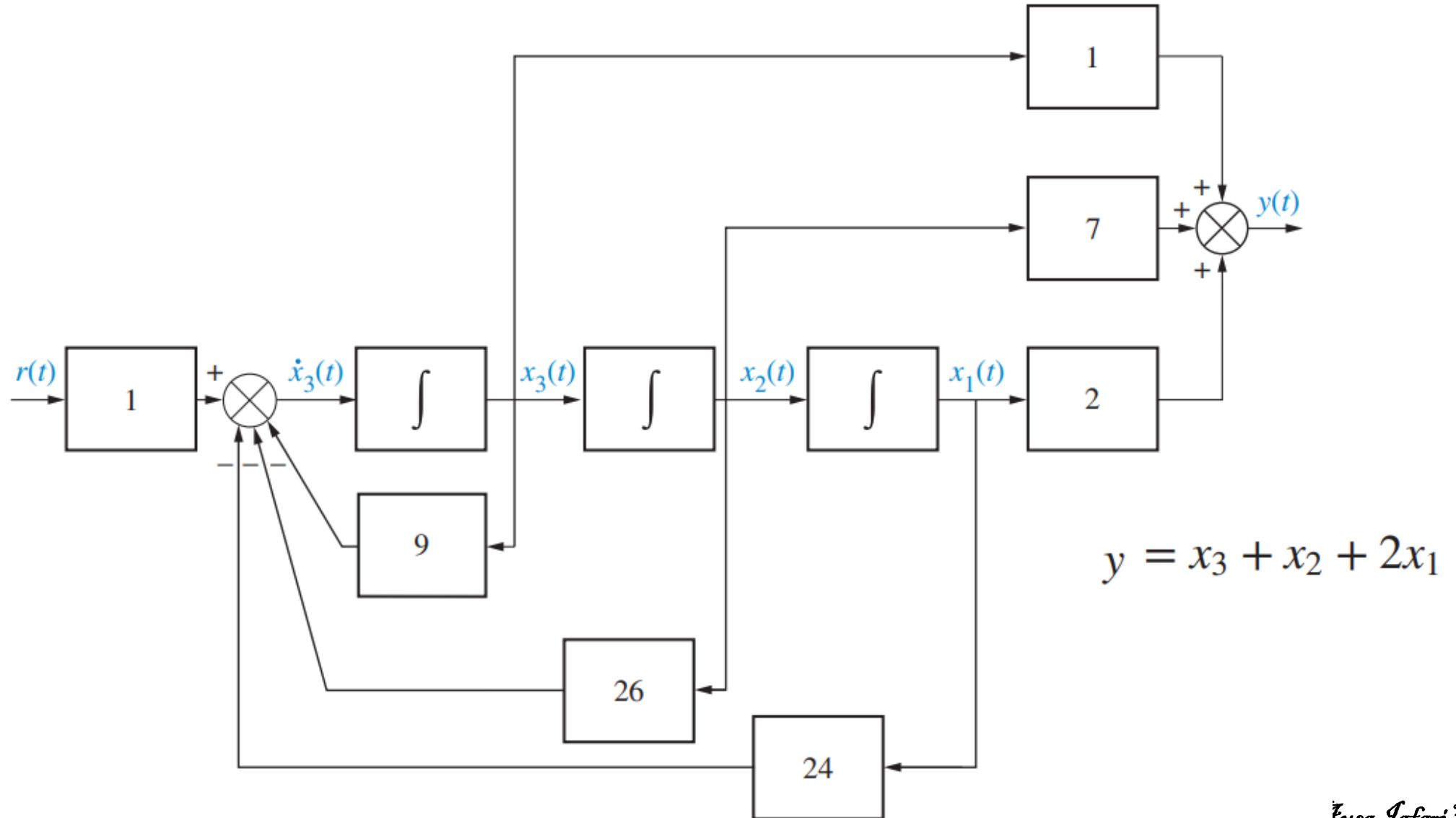
$$\ddot{x}_1 = x_3$$

$$y = c(t) = b_2 x_3 + b_1 x_2 + b_0 x_1 = x_3 + x_2 + 2x_1$$

$$y = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Converting a Transfer Function to State Space

E.g. (Cont.) Converting a Transfer Function with a Polynomial in the Numerator



Converting from State Space to a Transfer Function

Given the state and output equations:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

Take the Laplace transform assuming zero initial conditions

$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$

Solve for $\mathbf{X}(s)$

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}\mathbf{U}(s)$$

or

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s)$$

Converting from State Space to a Transfer Function

Substituting

$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) + \mathbf{D}\mathbf{U}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{U}(s)$$

The transfer function is achieved as:

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}$$

Converting from State Space to a Transfer Function

E.g.

PROBLEM: Given the system defined by Eq. (3.74), find the transfer function, $T(s) = Y(s)/U(s)$, where $U(s)$ is the input and $Y(s)$ is the output.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \quad (3.74a)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x} \quad (3.74b)$$

Sln.

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s+3 \end{bmatrix}$$

Converting from State Space to a Transfer Function

Sln. (Cont.)

Now form $(s\mathbf{I} - \mathbf{A})^{-1}$:

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} = \frac{\begin{bmatrix} (s^2 + 3s + 2) & s + 3 & 1 \\ -1 & s(s + 3) & s \\ -s & -(2s + 1) & s^2 \end{bmatrix}}{s^3 + 3s^2 + 2s + 1}$$

$$\mathbf{B} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{D} = 0$$

$$T(s) = \frac{10(s^2 + 3s + 2)}{s^3 + 3s^2 + 2s + 1}$$